Narrative Quantification:
Compiling Narrative Structures into Computable Representations for Large Cognitive Models

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Abstract

Human societies interpret events primarily through narrative structures rather than isolated data points. This paper proposes **Narrative Quantification**, a framework for transforming narrative information into structured cognitive data. By decomposing narratives into actors, events, conflicts, resolutions, emotional context, and causal relationships, narratives can be stored as structured datasets and analyzed computationally.

To operationalize this transformation, the paper introduces the concept of a **Narrative Compiler**, a system that converts narrative text into structured cognitive representations such as the **24TAG structure**. These representations enable the creation of narrative databases that may serve as the foundation for **Large Cognitive Models (LCM)**—systems designed to reason over structured representations of human narratives rather than purely probabilistic language tokens.

This work suggests that narrative quantification may function as a new layer of cognitive infrastructure for future AI systems.

Keywords: Narrative Quantification, Large Cognitive Models, 24TAG, Narrative Compiler, Cognitive Infrastructure, State Hash, JSON Canonicalization Scheme

1 Introduction

Most contemporary AI systems rely on probabilistic language modeling. Large Language Models (LLMs) generate text by predicting token probabilities within massive corpora. While effective for language generation, these systems do not explicitly represent narrative structure, causal relationships, or long-term narrative dynamics.

Human cognition, however, rarely processes information as isolated tokens. Instead, humans organize knowledge through **narratives**: structured relationships between actors, events, motivations, and outcomes.

This paper proposes that quantifying narrative structures can provide an alternative computational representation of human cognition. By decomposing narratives into structured components, it becomes possible to construct databases of collective narrative patterns.

2 Narrative Cognition

Human cognition frequently follows a narrative processing loop:

$$\text{Event} \rightarrow \text{Meaning} \rightarrow \text{Narrative} \rightarrow \text{Memory} \tag{1}$$

Experiences are rarely stored as raw data. Instead, individuals interpret events through causal explanations and emotional context, forming narratives that are later stored in memory. This mechanism allows humans to compress complex experiences into structured stories that **predict future outcomes and guide subsequent decisions**. Narrative cognition therefore represents a fundamental cognitive process rather than a mere cultural artifact.

3 Narrative Quantification

Narrative Quantification is the process of converting narrative structures into analyzable data. Narratives can be decomposed into structural components such as:

- Actors
- Events
- Conflict
- Resolution
- Emotional context
- Causal relationships

Once extracted, these components can be represented through structured tags and stored within databases. This enables large-scale analysis of narrative dynamics across media, politics, economics, and culture.

3.1 Decomposition

The first stage of quantification is **Decomposition**, where raw narrative text is segmented into its primary semantic units. Rather than treating a story as a continuous stream of words, we identify discrete entities and their roles. By isolating **Actors** (the agents of change), **Events** (the actions taken), and **Causal Relationships** (the 'why' behind the actions), we transform linguistic ambiguity into a set of distinct variables.

3.2 Event Formation

The second stage of quantification is **Event Formation**, where continuous occurrences are discretized into logical structures. Within this framework, **an event is defined as the minimum causal interaction unit between actors unfolding over time.**

Unlike traditional data processing, event formation recognizes the inherent unity of an action and its subsequent response. A typical event unit is structured as follows:

- **Actors:** The agents initiating or receiving the interaction.
- **Action-Reaction Pair:** The core causal loop.
- **Temporal Boundary:** The specific window from trigger to outcome.

By defining the event as the *minimum causal interaction unit*, we enable LCMs to perform reasoning not just on what happened, but on the **logic of the interaction itself**.

3.2.1 Eventualization: Entropy Suppression through Gravitational Convergence

Without event formation, information entropy increases monotonically over time. A single real-world occurrence typically generates dozens of articles across different media outlets — each carrying its own framing, emphasis, and omissions. Left unstructured, this proliferation of fragments causes the informational representation of a single event to *diverge*, consuming disproportionate computational and cognitive resources.

Eventualization is the process by which the **Event Builder** counteracts this divergence. Rather than treating each article as an independent data point, the Event Builder applies a gravitational model: semantically related articles are attracted toward a shared `event_id`, collapsing the high-entropy cloud of fragments into a single, stable coordinate in Narrative Space.

Formally, given a set of n articles $\{a_1, a_2, \ldots, a_n\}$ referencing the same real-world occurrence, Eventualization assigns them to a common event object E:

$$E = \text{EventBuilder}(\{a_1, \ldots, a_n\}), \quad \text{sim}(e(a_i), e(a_j)) \geq \theta \tag{2}$$

where $e(\cdot)$ is the embedding function mapping an article to its vector representation in Narrative Space, and θ is the similarity threshold above which two articles are considered to reference the same event. This operation is the narrative analogue of entropy reduction: the n fragments, each carrying partial and often contradictory information, are compressed into one canonical object whose informational content is greater than the sum of its parts.

3.2.2 The Centroid Vector and Crystallization Completeness

A direct consequence of Eventualization is the emergence of the **centroid vector** $\bar{v}$. Each article a_i assigned to event E contributes a 6-dimensional `strategic_interest_vector` $v_i \in [-1, 1]^6$, representing its narrative perspective across the dimensions of security, economy, technology, resource, ideology, and environment. The centroid is defined as the arithmetic mean across all contributing perspectives:

$$\bar{v} = \frac{1}{n} \sum_{i=1}^{n} v_i \tag{3}$$

The centroid represents the **gravitational center of the event** — the best available approximation of the event's objective geopolitical weight, computed from the ensemble of all contributing narrative perspectives. As more articles converge on the same `event_id`, the centroid stabilizes: the variance of $\{v_i\}$ decreases and the estimate of $\bar{v}$ becomes progressively more reliable.

This gives rise to a natural measure of **crystallization completeness**. When all six canonical narrative perspectives (jp, us, cn, gb, eu, qa) have been compiled for a given event, the `source_count` reaches 6. At this point the event is considered *fully crystallized* — its centroid has been computed from the maximum available set of viewpoints and no further perspective shift is expected. The distance between any individual narrative vector v_i and the centroid $\bar{v}$,

$$\text{CDC}_i = \|v_i - \bar{v}\|_2 \tag{4}$$

then functions as the **Cross-Domain Correlation (CDC)** metric: a quantitative measure of how far a given national narrative deviates from the ensemble center of gravity. High CDC values indicate narrative outliers — perspectives that are structurally distant from the consensus — and serve as the primary signal for the `audit_aura` indicator in the presentation layer.

3.3 Narrative Compression

The final stage of quantification is **Narrative Compression**, the process by which structured narrative data is refined into a high-density cognitive representation. This operation can be understood as an **active entropy reduction** within the information space.

Raw, real-world data is inherently high-entropy—it is noisy, continuous, and lacks explicit meaning. Narrative compression provides several key advantages:

- **Computational Efficiency:** Reasoning over discrete, compressed event units is far more efficient than processing raw linguistic tokens.

- **Long-Term Coherence:** Stripping away noise reveals the narrative arc, enabling tracking of long-term dynamics that LLMs often lose.
- **Storage and Retrieval:** Compression enables construction of vast, searchable Narrative Databases.

Ultimately, narrative compression transforms "text" into "knowledge." This compressed, structured format serves as the fundamental input for the Large Cognitive Model (LCM).

4 The Narrative Compiler: Search and Optimization

The Narrative Compiler does not merely extract data; it explores a vast space of potential interpretations to find the most probable narrative structures. Given that a single text can yield an exponential number of event connections, the Compiler employs specific **Search Constraints** to manage this complexity and generate the final probability distribution $P(N \mid \text{text})$.

4.1 Generating Possible Narrative Graphs

The Compiler treats narrative construction as a graph-search problem. From a set of extracted events, it attempts to build a **Narrative Graph** G where nodes represent events and edges represent causal or temporal transitions.

4.2 Beam Search for Hypothesis Management

To maintain a manageable "hash set" of narrative states, the Compiler utilizes **Beam Search**. Rather than exploring every possible graph configuration, the Compiler keeps only the top-k most plausible narrative hypotheses at each step of the construction.

- **Top-k Survival:** At each stage of event linking, only the graphs with the highest cumulative probability scores are retained.
- **Diversity Maintenance:** The beam search is tuned to preserve divergent narrative paths if their probabilities are sufficiently high.

4.3 Graph Pruning and Information Entropy

Graph Pruning is the active removal of low-probability or redundant branches in the narrative tree.

- **Redundancy Filtering:** Multiple paths leading to the same structural state with lower efficiency are pruned.
- **Entropy Thresholding:** Branches that increase conditional entropy without adding semantic value are eliminated.

4.4 Causal Constraints: The Logical Filter

The most critical constraint is the **Causal Constraint**, which acts as the "physics" of the Narrative Space. The Compiler applies logical rules to validate every proposed edge in the graph:

- **Temporal Precedence:** A cause must strictly precede its effect.
- **Actor Consistency:** Actions must align with the capabilities and known motivations of the involved actors.
- **Interaction Logic:** The Action-Reaction loop must maintain coherent flow.

4.5 Final State Synthesis

Through these constraints, the Narrative Compiler collapses the near-infinite possibilities of prose into a refined **hash set**. This set represents the "State" of the narrative—a weighted distribution of graphs that have survived beam search and passed all causal filters.

5 The 24TAG Taxonomy

To ensure universal compatibility and deterministic hashing, the Narrative Quantification protocol defines a fixed schema of 24 structured tags. These tags are categorized into four primary layers.

Table 1: The 24TAG Framework Definitions

Layer	Tag ID	Tag Name	Description
Actor & Agency	T01	Primary_Actor	Main entity initiating the interaction
	T02	Secondary_Actor	Entity receiving or responding to action
	T03	Actor_Role_A	Functional role of primary actor
	T04	Actor_Role_B	Functional role of secondary actor
	T05	Actor_Motivation_A	Underlying goal for Actor A
	T06	Actor_Motivation_B	Underlying goal for Actor B
Event & Action	T07	Action_Type	Category of event (Verbal, Physical, etc.)
	T08	Action_Intensity	Magnitude of action (0.0–1.0)
	T09	Target_Resource	Specific object or value being contested
	T10	Event_Modality	Factual, hypothetical, or desired
	T11	Temporal_Sequence	Relative order within narrative arc
	T12	Spatial_Context	Domain or environment of event
Causal & Logic	T13	Causal_Link_Type	Nature of connection (Direct Cause, etc.)
	T14	Conflict_Nature	Type of opposition (Internal, etc.)
	T15	Conflict_Intensity	Severity of friction between actors
	T16	Resolution_Status	Degree of conflict settlement
	T17	Outcome_Valence	Positive or negative impact
	T18	Logic_Consistency	Alignment with prior narrative states
Context & Tone	T19	Emotional_Tone_A	Emotional state of Actor A
	T20	Emotional_Tone_B	Emotional state of Actor B
	T21	Perspective_Bias	Point of view of the narrative
	T22	Info_Asymmetry	Knowledge gap between actors
	T23	Significance_Score	Relative importance of event
	T24	Narrative_Closure	Extent of thread conclusion

5.1 Data Structure and Compilation

Each tag ($T_{01...24}$) is populated by the Narrative Compiler using the Beam Search and Causal Constraints defined in Section 4. The resulting data structure is a JSON object. As specified in the **Crystallization Protocol**, the `state_hash` is calculated by applying SHA-256 to the JCS-normalized string of these 24 tags:

```
{
  "T01": "Actor_A",
  "T02": "Actor_B",
  ...
  "T24": "Closure_Value",
  "state_hash": "6b86b273ff34fce19d6b804eff5a3f57..."
}
```

- **Normalization (JCS):** Following RFC 8785 [1], the Compiler sorts the 24 structured tags into a canonical JSON format.
- **Self-Referential Exclusion:** The `state_hash` field itself is strictly excluded from the hash input to prevent circular dependencies.
- **Hashing (SHA-256):** The Compiler applies SHA-256 to the canonical byte-stream of the 24 tags.

The mathematical definition of the state is:

$$state_hash = \text{SHA256}(\text{JCS}(\{TAG_{01...24}\} \setminus \{state_hash\})) \tag{5}$$

5.2 Integrity and Verification

By excluding the `state_hash` from its own input, we create a **One-Way Cognitive Seal**. A receiver can verify the integrity of a narrative state by re-running the JCS and SHA-256 process on the remaining 24 tags and comparing against the original `state_hash`.

6 Narrative Observation and the Narrative Space

6.1 Defining Narrative Space

Definition 1. ***Narrative Space*** *is the multi-dimensional space formed by the set of possible narrative states. Each specific state is represented as a point within this space.*

6.2 The Mathematical Foundation: Graph Embedding

Each narrative graph G is projected into a d-dimensional real-valued space via an embedding function e:

$$e : G \to \mathbb{R}^d \tag{6}$$

6.3 Quantifying Divergence through Distance

The divergence between two narrative states G_1 and G_2 is defined by the distance between their corresponding embeddings:

$$\text{distance} = \|e(G_1) - e(G_2)\| \tag{7}$$

- **Low Distance (Convergence):** Indicates cognitive consensus.
- **High Distance (Divergence):** Indicates competing interpretations or "Narrative Rifts."

6.4 Mapping the Narrative Ecosystem

By observing the distribution and trajectory of vectors $e(G)$ within the Narrative Space, researchers can identify:

- **Attractors:** Regions where narrative vectors cluster, indicating dominant societal beliefs.
- **Bifurcation Points:** Moments where a single event causes the distribution to split into two distant clusters.
- **Narrative Velocity:** The rate at which $e(G)$ moves through the space as new information is integrated.

7 Cognitive Infrastructure

7.1 From Knowledge Bases to Narrative Databases

Traditional knowledge bases (e.g., Wikidata, Knowledge Graphs) are effective at storing *what* happened—discrete facts such as dates, locations, and actors. However, they lack the capacity to store *how* those facts are interpreted. A **Narrative Database**, powered by the Narrative Compiler, stores **Crystallized Narrative States**. This shifts the focus from "Data Storage" to "Cognitive State Storage."

7.2 Information Integrity and the State Hash

The deterministic nature of the `state_hash` provides a mechanism for **Information Integrity**:

- **Proof of Interpretation:** A `state_hash` attached to a report provides a "Cognitive Seal."
- **Divergence Detection:** Comparing `state_hash` outputs across media outlets objectively indicates narrative structural differences.

7.3 Applications of the Infrastructure

- **Media & Information Integrity:** Automated detection of narrative manipulation.
- **Geopolitical Narrative Mapping:** Real-time observation of how different nations interpret the same international event.
- **Corporate & Economic Intelligence:** Tracking the evolution of market narratives through structured event formation.

8 Reasoning over Narrative Distributions (LCM)

By defining a state as $P(N \mid \text{text})$, the **Large Cognitive Model (LCM)** gains the ability to perform "superpositional" reasoning.

8.1 Inference as Distribution Transformation

While LLMs predict tokens, the LCM performs inference by transforming one probability distribution into another. The reasoning core evaluates how the current distribution $P(N_t \mid \text{text}_t)$ evolves into a future distribution $P(N_{t+1})$ when a new event occurs.

8.2 Handling Ambiguity and Divergence

This probabilistic approach allows the LCM to handle **Narrative Divergence** natively. When a text is highly ambiguous, the Compiler produces a broad "hash set" with distributed probabilities. The LCM can track multiple "possible worlds" simultaneously.

8.3 Reasoning over Causal Trajectories

Because an event is defined as the *minimum causal interaction unit*, the LCM can track long-term narrative arcs without the "forgetting" or "hallucination" common in LLMs. Reasoning in an LCM is not a search for the next word, but a calculation of **state transitions**:

$$\text{State } A\,(\text{Hash}_1) \xrightarrow{\text{Causal Logic}} \text{State } B\,(\text{Hash}_2) \tag{8}$$

9 Human Memory and Narrative Cognition

Human memory does not function as a raw sensory log; rather, it operates through a sophisticated process of **narrative compression**.

9.1 Memory as a Compiled Distribution

When humans experience an event, the brain acts as a biological **Narrative Compiler**, transforming continuous experience into a discrete set of structured narrative units. Human memory can be viewed as the storage of the probability distribution:

$$P(N \mid \text{Experience}) \tag{9}$$

What we "remember" is not the raw text of reality, but the **hash set** of the most probable narrative structures.

9.2 The Primitives of Biological Storage

The human cognitive system captures only essential semantic primitives:

- **Key Actors:** Identifying who was involved.
- **Causal Interpretations:** Assigning a "why" to the sequence of events.
- **Emotional Evaluation:** Tagging the event with a valence.
- **Outcomes:** The final state after the interaction is resolved.

9.3 Biomimetic Validation

The approach of Narrative Quantification is fundamentally **biomimetic**. By using the `state_hash` to represent a "crystallized" interpretation, we emulate the way human synapses consolidate complex experiences into stable, retrievable "concepts."

10 Limitations

10.1 Subjectivity and Algorithmic Bias

The primary challenge is the inherent subjectivity of human interpretation. The "priors" used to assign probabilities are often derived from the training data of the underlying models, which can introduce systemic biases where the compiler favors specific cultural tropes.

10.2 Computational Complexity of Graph Generation

Constructing **possible narrative graphs** is a non-trivial computational task. In complex, multi-actor texts, the number of potential causal connections grows exponentially.

10.3 Cultural and Linguistic Heterogeneity

Narrative structures are not universal. Different cultures employ different storytelling archetypes. A 24TAG framework optimized for one may fail to capture the nuances of another.

10.4 Data Sovereignty and "Narrative Warfare"

If a centralized authority can define the "standard" `state_hash` for a global event, they possess the power to delegitimize divergent but valid interpretations. The risk of using this infrastructure for "narrative warfare" is a serious ethical concern.

10.5 Semantic Loss during Compression

Narrative compression is an active entropy reduction process. There is a persistent risk that subtle but crucial "soft" data—such as irony, subtext, or poetic ambiguity—may be lost during the transition from raw text to a structured `state_hash`.

11 Conclusion

Despite these limitations, Narrative Quantification represents a necessary evolution in Artificial Intelligence. By moving beyond the probabilistic prediction of language tokens and toward the **structured reasoning of narrative states**, we provide a framework that more closely mirrors human cognition.

The transition from **Narrative Structures to Large Cognitive Models (LCM)** offers more than just technical efficiency; it provides a way to map the "Narrative Space" of our society, making the invisible structures of our interpretations visible, measurable, and verifiable.

Future research will focus on mitigating bias through diverse "compiler ensembles" and exploring the decentralized governance of narrative hashes to ensure that this cognitive infrastructure serves as a tool for clarity, not control.

The path toward **Artificial General Intelligence (AGI)** lies not in bigger models, but in deeper structures. Narrative Quantification is the first step in building an AI that doesn't just talk like a human, but understands like one.

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